

PHYSICS • UPPER SIXTH

UPPER SIXTH — PHYSICS

## CHAPTER 11

# THE ATOM

SUGGESTED TIME: 3 PERIODS

### IN THIS CHAPTER

This chapter explores the structure of the atom, from Rutherford's nuclear model revealed by alpha-particle scattering to the quantised energy levels that explain atomic emission and absorption spectra. It introduces the electron volt, the photon relation  $E = hf = E_2 - E_1$ , and the Schrodinger model of the hydrogen atom, which are essential tools for the GCE Advanced Level examination.

- By the end of this chapter I can describe the historical atomic models (Dalton, Thomson, Rutherford, Bohr) and state their limitations
- By the end of this chapter I can describe Rutherford's alpha scattering experiment, its observations and the conclusions drawn about the nucleus
- By the end of this chapter I can estimate the size of the nucleus from scattering data and explain why most alpha particles pass through undeflected
- By the end of this chapter I can distinguish continuous, line emission and line absorption spectra and explain their origin
- By the end of this chapter I can explain energy levels, ground state, excited states and ionisation energy of an atom
- By the end of this chapter I can define and use the electron volt and convert between eV and joules



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## CHAPTER OBJECTIVES

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- By the end of this chapter I can define and use the electron volt and convert between eV and joules
- By the end of this chapter I can apply  $E = hf = E_2 - E_1$  to calculate photon frequencies, wavelengths and energies in transitions
- By the end of this chapter I can outline the Schrodinger (quantum mechanical) model of the hydrogen atom and the meaning of quantum numbers
- By the end of this chapter I can interpret a hydrogen energy level diagram and identify the Lyman, Balmer and Paschen series (GCE exam extra)

**PREREQUISITES**

- Wave nature of light: frequency, wavelength and the relation  $c = f \lambda$  (Lower Sixth)
- Photoelectric effect and the photon concept  $E = hf$  (Lower Sixth)
- Electric charge, Coulomb force and electric potential energy
- Kinetic energy and conservation of energy
- Orders of magnitude and use of scientific notation with SI units

## 1 ATOMIC MODELS AND RUTHERFORD'S ALPHA SCATTERING EXPERIMENT

When a welder in Douala strikes an arc, the brilliant blue-white light hurts the eyes, yet the sodium street lamps of Yaoundé glow a soft orange, and the neon signs of Bamenda shine red. Why does each gas emit its own characteristic colours, and why does white sunlight, spread out by a CD or a prism, show a continuous rainbow while a gas discharge tube shows only a few sharp coloured lines? Even more puzzling: why does the negatively charged electron not simply spiral into the positive nucleus, as classical physics predicts? The answers lie inside the atom itself, and they were uncovered by bombarding gold foil with alpha particles and by analysing the light atoms emit and absorb. This chapter takes you on that journey, from Rutherford's scattering experiment to the quantum picture of the hydrogen atom.

Before we can explain the light, we must know what the atom looks like. That story begins with a simple question: is matter infinitely divisible, or is there a smallest piece?

### 1.1 A brief history of atomic models

#### Dalton's indivisible sphere (early 19th century).

John Dalton, an English schoolteacher, proposed that each chemical element is made of identical, tiny, **indivisible** solid spheres called atoms. Atoms of different elements differ in mass, and chemical reactions simply rearrange atoms without creating or destroying them. This model explained the laws of chemical combination beautifully, but it said nothing about the internal structure of the atom — for Dalton, there was none.

#### Thomson and the discovery of the electron (1897).

Studying cathode rays in evacuated discharge tubes, J. J. Thomson showed that the rays consist of negatively charged particles — **electrons** — far lighter than any atom, and that they are the same whatever gas or metal is used. The electron is therefore a constituent of every atom: Dalton's indivisible sphere had been divided.

Since atoms are electrically neutral, the negative electrons must be balanced by positive charge somewhere. Thomson proposed the **plum-pudding model**: the atom is a sphere of uniformly distributed positive charge (the "pudding"), with electrons embedded in it like plums. The model is neat, but it was never tested directly — until Ernest Rutherford and his students aimed alpha particles at gold foil.

#### **WARNING** Two models, one test

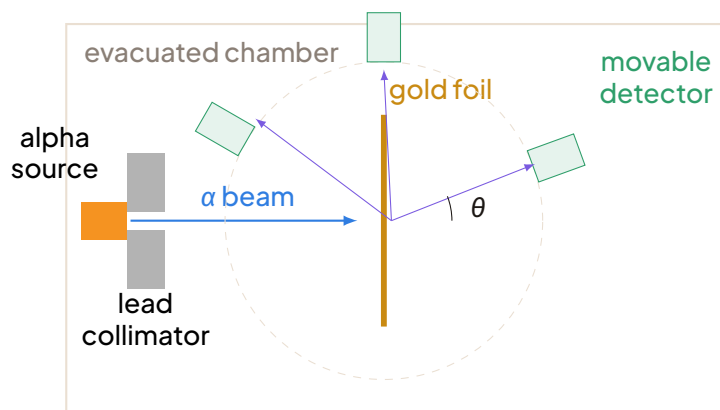
Dalton's atom is a hard, featureless ball; Thomson's atom is a soft, diffuse ball of positive charge with electrons inside. Both predict that a fast, heavy, positively charged projectile fired at a thin foil should pass through with only tiny deflections, because no concentrated charge exists to push it hard. Rutherford's experiment was designed to check exactly this prediction.

## 1.2 Rutherford's alpha scattering experiment (Geiger and Marsden, 1909)

### EXPERIMENT The gold foil experiment

In Rutherford's laboratory in Manchester, Hans Geiger and Ernest Marsden set up the following apparatus (see figure below):

- a radioactive source (radon) emitting **alpha particles** — helium nuclei  ${}^4_2\text{He}$ , charge  $+2e$ , mass about  $6.6 \times 10^{-27} \text{ kg}$  — at speeds of about  $2 \times 10^7 \text{ m s}^{-1}$ ;
- a thick lead block with a narrow hole, acting as a collimator to produce a fine beam;
- an extremely thin gold foil (about  $1 \times 10^{-6} \text{ m}$  thick, only a few thousand atoms) as target; gold is used because it can be beaten exceptionally thin and has a large atomic number  $Z = 79$ ;
- a scintillation detector (a zinc sulphide screen viewed through a microscope): each alpha particle striking it produces a tiny flash of light. The detector can be moved around the foil to count particles at any angle  $\theta$ ;
- the whole arrangement sits in an evacuated chamber, so that the alpha particles are not scattered by air molecules.



Rutherford's scattering apparatus: a collimated beam of alpha particles strikes a thin gold foil inside an evacuated chamber; the scintillation detector counts particles at each scattering angle  $\theta$ .

## 1.3 Observations

Counting the flashes at many angles, Geiger and Marsden found three striking facts:

1. The **vast majority** of alpha particles pass straight through the foil with little or no deflection ( $\theta$  less than about  $1^\circ$ ).
2. A **small fraction** are deflected through large angles.

3. About 1 in 8000 alpha particles is scattered through more than  $90^\circ$  — it effectively bounces back towards the source.

Rutherford later said this was as surprising as firing a shell at a piece of tissue paper and having it come back and hit you. No diffuse “pudding” of positive charge can reverse the motion of a fast, heavy alpha particle; only a very concentrated charge can.

## 1.4 Conclusions: the nuclear atom

### DEFINITION The atomic nucleus

The **nucleus** is the tiny, dense, positively charged centre of the atom. It contains nearly all the atom’s mass and all its positive charge  $+Ze$  (where  $Z$  is the atomic number), concentrated in a region of radius about  $10^{-15}$  m to  $10^{-14}$  m, while the whole atom has a radius of about  $10^{-10}$  m. The light, negatively charged electrons occupy the enormous, almost empty space around it. The nucleus is composed of **protons** (charge  $+e$ ) and **neutrons** (charge 0), collectively called **nucleons**.

The three observations translate directly into three conclusions:

- Most particles pass through undeflected  $\Rightarrow$  the atom is **mostly empty space**.
- Some particles are strongly deflected  $\Rightarrow$  the positive charge is concentrated in a **very small region**, so that a close encounter produces a huge Coulomb force.
- A few particles bounce back  $\Rightarrow$  nearly all the **mass** is in that same tiny region, so the alpha particle (much lighter than a gold nucleus) rebounds.

### WARNING Atom versus nucleus: a factor of $10^4$

A classic examination trap is to confuse the radius of the atom ( $\approx 10^{-10}$  m) with that of the nucleus ( $\approx 10^{-14}$  m). If the nucleus were the size of a marble placed at the centre of the Ahmadou Ahidjo stadium, the electrons would be moving around the furthest stands — everything in between is empty space.

## 1.5 Qualitative explanation of the scattering

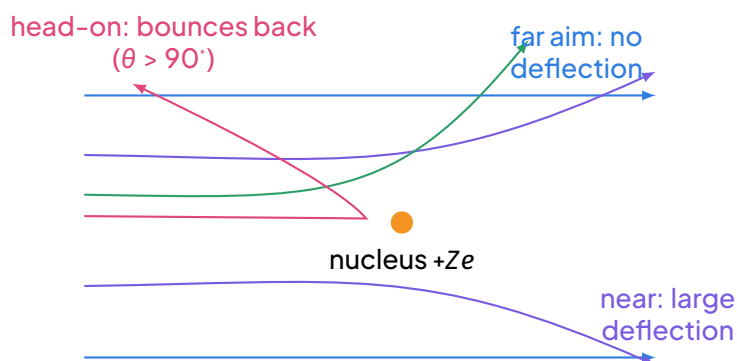
The alpha particle carries charge  $+2e$  and the nucleus carries charge  $+Ze$ . Between them acts the **Coulomb repulsion**

$$F = \frac{1}{4\pi\epsilon_0} \frac{(2e)(Ze)}{r^2},$$

which grows rapidly as the separation  $r$  decreases. An alpha particle aimed far from a nucleus feels only a weak, brief push and continues almost straight. One aimed close to a nucleus feels a strong force for a longer time and is deflected through a large angle. One aimed almost head-on is slowed, stopped momentarily, and repelled straight back — this is the rare  $\theta > 90^\circ$  event.

**WARNING** Repulsion, not attraction

A very common mistake is to say that alpha particles are attracted by the nucleus. Both are positively charged, so the force is repulsive. The curved trajectories bend away from the nucleus, never towards it.



Alpha particle trajectories near a nucleus. The closer the line of approach to the nucleus, the stronger the Coulomb repulsion and the larger the deflection; a nearly head-on particle is scattered backwards.

**PROPERTY** Factors affecting large-angle scattering

The probability of large-angle scattering:

- increases with the thickness of the foil (more nuclei on the path);
- increases with the atomic number  $Z$  of the target (stronger repulsion);
- decreases as the kinetic energy of the alpha particles increases (faster particles spend less time near the nucleus and are harder to deflect).

This qualitative dependence is examinable; the full derivation of Rutherford's scattering formula is not required.

## 1.6 Estimating the size of the nucleus

Consider an alpha particle fired directly at a nucleus of charge  $+Ze$ . Far away it has kinetic energy  $E_k$ . As it approaches, the Coulomb repulsion does negative work on it: it slows down, converting kinetic energy into electric potential energy. At the **distance of closest approach**  $r_0$ , the particle is momentarily at rest, so all its initial kinetic energy has become potential energy.

**METHOD** Estimating the nuclear radius from closest approach

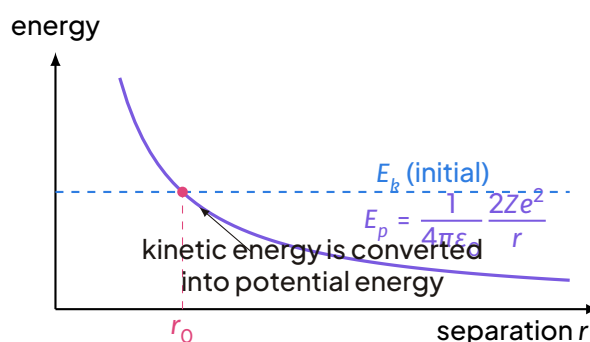
1. Write the initial kinetic energy of the alpha particle:  $E_k = \frac{1}{2}m_\alpha v^2$ .
2. Write the electric potential energy of the pair (alpha  $+2e$ , nucleus  $+Ze$ ) at separation  $r$ :

$$E_p = \frac{1}{4\pi\epsilon_0} \frac{(2e)(Ze)}{r}.$$

3. At the distance of closest approach, apply conservation of energy:  $E_k = E_p(r_0)$ .
4. Solve for  $r_0$ :

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{E_k}.$$

5. Interpret:  $r_0$  is an upper limit for the nuclear radius, since the particle stops before (or just at) the nuclear surface; if it penetrated the nucleus, the strong nuclear force would alter the interaction.



Energy diagram for a head-on collision. As the alpha particle approaches, its kinetic energy falls while the electric potential energy rises; at  $r = r_0$  the particle stops momentarily and  $E_k = E_p$ .

**WORKED EXAMPLE 1** Upper limit for the radius of a gold nucleus

An alpha particle of kinetic energy  $E_k = 5.0 \text{ MeV}$  is fired head-on at a gold nucleus ( $Z = 79$ ). Estimate the distance of closest approach and hence an upper limit for the radius of the gold nucleus.

Data:  $e = 1.60 \times 10^{-19} \text{ C}$ ,  $\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ ,  $1 \text{ MeV} = 1.60 \times 10^{-13} \text{ J}$ .

**SOLUTION**

Step 1 — Convert the energy to joules.

$$E_k = 5.0 \times 1.60 \times 10^{-13} = 8.0 \times 10^{-13} \text{ J.}$$

Step 2 — Apply energy conservation at closest approach.

$$E_k = \frac{1}{4\pi\epsilon_0} \frac{(2e)(Ze)}{r_0} \implies r_0 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{E_k}.$$

Step 3 — Substitute.

$$r_0 = \frac{8.99 \times 10^9 \times 2 \times 79 \times (1.60 \times 10^{-19})^2}{8.0 \times 10^{-13}} = \frac{8.99 \times 10^9 \times 158 \times 2.56 \times 10^{-38}}{8.0 \times 10^{-13}}.$$

$$r_0 = \frac{3.64 \times 10^{-26}}{8.0 \times 10^{-13}} \approx 4.5 \times 10^{-14} \text{ m.}$$

Conclusion. The gold nucleus has a radius smaller than about  $4.5 \times 10^{-14} \text{ m}$  — roughly  $10^4$  times smaller than the atom ( $\approx 10^{-10} \text{ m}$ ). This confirms that the atom is overwhelmingly empty space.

## 1.7 Rutherford's planetary model — and its failure

To complete the picture, Rutherford proposed the **nuclear (planetary) model**: the electrons orbit the nucleus as planets orbit the Sun, the Coulomb attraction providing the centripetal force:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{m_e v^2}{r}.$$

The model explains the scattering data perfectly, but it contains a fatal flaw. According to classical electromagnetism, any accelerated charge radiates energy. An orbiting electron is permanently accelerated (centripetal acceleration), so it should continuously emit electromagnetic radiation, lose energy, and **spiral into the nucleus** in a tiny fraction of a second.

### **WARNING** Two failures of the planetary model

- **Stability:** the radiating electron should collapse into the nucleus — yet atoms are stable.
- **Spectra:** a spiralling electron would emit a continuous range of frequencies — yet gases emit only a few sharp, discrete lines (the orange of sodium, the red of neon).

Classical physics cannot save the planetary atom. A new idea is needed: energy in the atom is quantised. That is the subject of the next sections.

**DID YOU KNOW?** From Manchester to your examination

Rutherford, a New Zealander working in Manchester, is often called the father of nuclear physics. His scattering technique — probing something too small to see by firing particles at it and studying how they bounce — is still the basic method of particle physics today, in machines such as the Large Hadron Collider, where protons replace alpha particles and gold nuclei are replaced by other protons.

**KEY POINTS** Key points of this section

- Dalton: indivisible spheres; Thomson: electrons embedded in a positive “pudding”.
- Rutherford’s experiment: alpha particles on thin gold foil in vacuum, detected by scintillations at all angles.
- Observations: most particles pass straight through; a few are deflected strongly; about 1 in 8000 bounces back.
- Conclusions: the atom is mostly empty space; a tiny, dense, positive **nucleus** ( $\approx 10^{-14}$  m) holds nearly all the mass; the atom is  $\approx 10^{-10}$  m across.
- Scattering is due to Coulomb repulsion between the positive alpha particle and the positive nucleus; closer approach means larger deflection.
- Nuclear size estimate: equate initial kinetic energy to potential energy at closest approach,  $E_k = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{r_0}$ .
- The planetary model fails: an orbiting electron should radiate and spiral into the nucleus; it cannot explain atomic stability or line spectra.

**EXERCISE 1** Checking your understanding

1. State the three main observations of the Geiger–Marsden experiment and the conclusion drawn from each.
2. Explain why the scattering chamber must be evacuated.
3. An alpha particle of kinetic energy 7.7 MeV approaches a silver nucleus ( $Z = 47$ ) head-on. Calculate the distance of closest approach. ( $1 \text{ MeV} = 1.60 \times 10^{-13} \text{ J}$ .)
4. Why would an aluminium foil ( $Z = 13$ ) of the same thickness produce fewer large-angle scatterings than the gold foil?

**SOLUTION** Exercise 1

1. Most alpha particles pass through undeflected  $\Rightarrow$  the atom is mostly empty space. A few are deflected through large angles  $\Rightarrow$  the positive charge is concentrated in a very small region. About 1 in 8000 rebounds  $\Rightarrow$  nearly all the mass is concentrated in that tiny nucleus.
2. Air molecules would scatter and absorb the alpha particles, masking the scattering by the foil itself.
3. 
$$r_0 = \frac{8.99 \times 10^9 \times 2 \times 47 \times (1.60 \times 10^{-19})^2}{7.7 \times 1.60 \times 10^{-13}} = \frac{2.16 \times 10^{-26}}{1.23 \times 10^{-12}} \approx 1.8 \times 10^{-14} \text{ m.}$$
4. The Coulomb repulsion is proportional to  $Z$ ; with a smaller  $Z$ , the force on the alpha particle is weaker, so large deflections are less probable.

## 2 EMISSION AND ABSORPTION SPECTRA

Rutherford's scattering experiment told us where the mass and positive charge of the atom sit: in a tiny, dense nucleus. But it said nothing about how the electrons are arranged around that nucleus, nor about the energy they carry. The next great clue did not come from bombarding atoms with particles, but from something far gentler: simply **looking carefully at the light that matter emits and absorbs**. When we spread light out into its component colours — when we produce a **spectrum** — matter reveals a hidden barcode that no chemical test can imitate. This section is about reading that barcode.

### 2.1 Producing a spectrum: dispersion and the spectrometer

**Observation.** Hold a glass prism in a sunbeam near a window in Yaounde on a bright afternoon and a rainbow appears on the wall. The white sunlight has been separated into its colours. This separation is called **dispersion**: the refractive index of glass depends slightly on wavelength, so each colour is deviated through a slightly different angle (violet most, red least).

A **diffraction grating** — a slide ruled with thousands of equally spaced parallel lines per millimetre — does the same job by interference rather than refraction, and usually spreads the colours out much more, allowing precise measurement of wavelengths using the grating equation  $d \sin \theta = n\lambda$ .

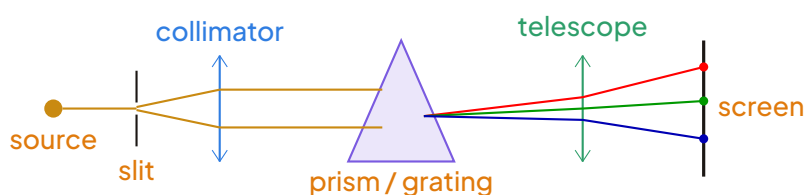
**DEFINITION** Spectrum and wavelength

A **spectrum** is the display obtained when light is separated into its component wavelengths. Each colour corresponds to a definite wavelength  $\lambda$  (in metres, often quoted in nanometres,  $1 \text{ nm} = 10^{-9} \text{ m}$ ) and frequency  $f$ , linked by

$$c = f\lambda, \quad c = 3.00 \times 10^8 \text{ m s}^{-1}.$$

To study spectra quantitatively we use a **spectrometer**, which has three essential parts:

- a **collimator**: a narrow slit placed at the focus of a converging lens, producing a parallel beam from the source;
- a **dispersing element**: a prism or a diffraction grating, which sends different wavelengths in different directions;
- a **telescope** (or screen/camera): a second lens system that focuses each direction to a sharp image of the slit — one image per wavelength present.



A simple prism spectrometer: the collimator makes a parallel beam, the prism disperses it, and the telescope focuses each wavelength to a separate line image on the screen.

## 2.2 The three types of spectra

When we point a spectrometer at different light sources, exactly **three** kinds of spectra appear. Learning to recognise them instantly is essential for the GCE.

**1. The continuous spectrum.** Heat a solid until it glows — the tungsten filament of an incandescent lamp, the iron of a blacksmith's forge in Douala — and its spectrum is an unbroken band of colour from red to violet. Every wavelength in a broad range is present.

### DEFINITION Continuous spectrum

A **continuous spectrum** contains all wavelengths over a broad range, with no gaps. It is emitted by hot solids, hot liquids, and dense (high-pressure) gases. The Sun's photosphere and a filament lamp are standard examples.

**2. The line emission spectrum.** Now replace the source by a **gas at low pressure** in a discharge tube (a glass tube with two electrodes, like a neon sign) or by a flame coloured by a metallic salt. The rainbow collapses: instead of a continuous band you see a few **bright coloured lines on a dark background**, each line being an image of the slit at one particular wavelength.

### DEFINITION Line spectrum

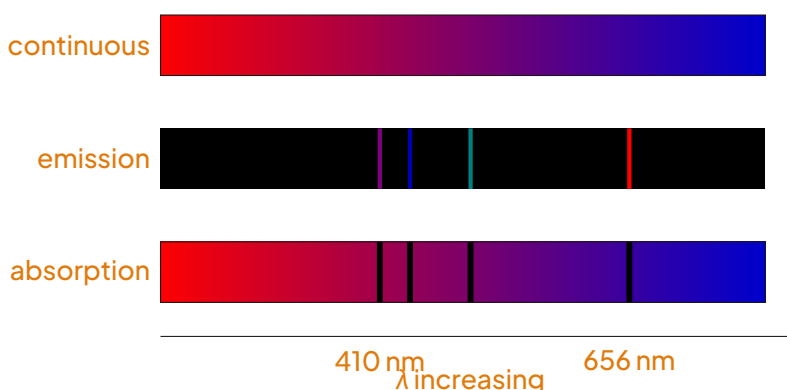
A **line spectrum** is a spectrum consisting of discrete bright (emission) or dark (absorption) lines at definite wavelengths, characteristic of an element.

Hydrogen, for example, shows in the visible region a red line ( $\lambda = 656 \text{ nm}$ ), a blue-green line (486 nm), and violet lines (434 nm, 410 nm) — and nothing in between. Sodium vapour is dominated by a brilliant yellow doublet near 589 nm, which is why sodium street lamps bathe everything in yellow.

**3. The line absorption spectrum.** Finally, let white light (which would give a continuous spectrum) pass through a **cooler gas** before entering the spectrometer. The continuous band reappears, but it is now crossed by dark lines — narrow wavelengths missing from the rainbow.

**PROPERTY** Kirchhoff's laws of spectroscopy

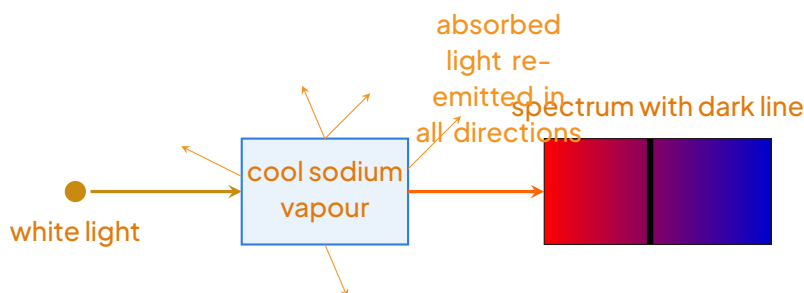
1. A hot solid, liquid, or dense gas produces a continuous spectrum.
2. A hot, low-pressure gas produces a line emission spectrum (bright lines on dark background).
3. A continuous spectrum source viewed through a cooler, low-pressure gas produces a line absorption spectrum (dark lines on continuous background). The dark lines occur at exactly the same wavelengths as the bright lines in the emission spectrum of the same gas.



The three types of spectra. Top: continuous spectrum of a hot solid. Middle: emission line spectrum of hydrogen (visible Balmer lines). Bottom: absorption spectrum of the same gas — the dark lines sit at exactly the same wavelengths as the bright emission lines.

### 2.3 Why absorption lines appear: a closer look

The absorption spectrum deserves careful thought, because it hides a subtlety that examiners love to test.



White light crossing cool sodium vapour. Atoms absorb strongly at their characteristic wavelengths and re-emit the energy in random directions, so the forward beam is weakened at those wavelengths: a dark line appears.

When white light traverses the cool gas, atoms absorb photons of their characteristic wavelengths very strongly. But the atoms do not keep this energy: a tiny fraction of a second later they re-emit it — **in random directions**. Almost none of the re-emitted light travels forward into the spectrometer, so the forward beam is depleted at exactly those wavelengths: a dark line. The gas has not "ignored" those wavelengths — quite the opposite, it has interacted with them more strongly than with any other.

**WARNING** Common mistake: misreading the dark lines

The dark absorption lines do not mean that the gas is transparent to those wavelengths or fails to interact with them. The gas absorbs those wavelengths very strongly and then re-emits the energy in all directions, so little of it continues along the original beam. Dark line = strong absorption, not absence of interaction.

**WARNING** Common mistake: confusing continuous and line spectra

A continuous spectrum contains every colour in a range (hot solid, filament lamp). A line spectrum contains only a few discrete wavelengths (low-pressure gas). Never describe a discharge-tube spectrum as "all the colours of the rainbow", and never say a filament lamp emits "lines".

## 2.4 Spectra as fingerprints of the elements

Here is the fact that turned spectroscopy into the most powerful identification tool in science: **no two elements have the same line spectrum**. Hydrogen's four visible lines, sodium's yellow doublet, mercury's green and violet lines — each element signs its light with a unique pattern.

**METHOD** Identifying an element from its spectrum

1. Produce the spectrum of the unknown sample (discharge tube, flame, or starlight through a spectrometer).
2. Measure the wavelengths of the bright lines (emission) or dark lines (absorption).
3. Compare these wavelengths with reference spectra tabulated for known elements.
4. A match of several line wavelengths identifies the element unambiguously — even if it is present only as a trace, or located in a star millions of kilometres away.

**DID YOU KNOW?** Helium: the element discovered in the Sun before the Earth

In 1868, astronomers studying the Sun's spectrum during an eclipse found a yellow line that matched no known element. They concluded the Sun contained a new substance and named it helium, from the Greek helios, "Sun". Helium was only found on Earth 27 years later. The dark lines in the Sun's spectrum — the **Fraunhofer lines** — are absorption lines produced when light from the hot photosphere crosses the cooler gases of the solar atmosphere; they let us determine the chemical composition of a star we can never visit.

**2.5 The electron volt (eV)**

In atomic physics, the joule is inconveniently large. We define a much smaller unit of energy: the **electron volt**.

**DEFINITION** Electron volt

One **electron volt (1 eV)** is the kinetic energy gained by an electron when it is accelerated through a potential difference of 1 V.

$$1\text{ eV} = e \times 1\text{ V} = 1.60 \times 10^{-19}\text{ C} \times 1\text{ J C}^{-1} = 1.60 \times 10^{-19}\text{ J}.$$

**KEY POINTS** Using the electron volt

- Atomic energy levels are typically a few eV (e.g. hydrogen ground state -13.6 eV).
- Photon energies in the visible range are 1.8 eV (red) to 3.1 eV (violet).
- Always convert to joules when using  $E = hf$  with  $h = 6.63 \times 10^{-34}\text{ J s}$ , or use  $h = 4.14 \times 10^{-15}\text{ eV s}$ .

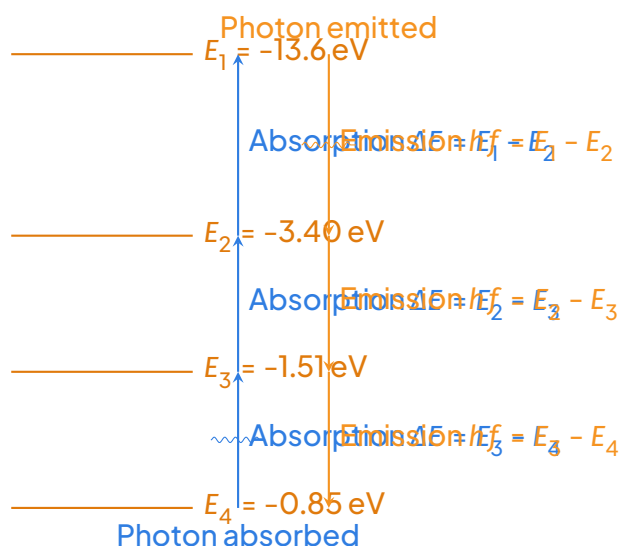
## 2.6 The qualitative link to the atom: energy is quantised

Now comes the deep question: why do atoms emit and absorb only certain frequencies? A hot solid emits everything; a rarefied gas of atoms emits a handful of precise notes, like an instrument that can play only a few keys.

The inescapable conclusion — which the next section develops quantitatively — is this:

### KEY POINTS The message of line spectra

Atoms emit or absorb light only at specific, well-defined frequencies. Since light of frequency  $f$  carries energy  $E = hf$ , this means an atom can only gain or lose definite amounts of energy. The internal energy of an atom is **quantised**: it exists only in certain allowed values called energy levels. Each spectral line corresponds to a transition between two such levels.



Energy level diagram for hydrogen (not to scale). Upward arrows (blue): absorption of a photon of exact energy  $\Delta E = hf$ . Downward arrows (orange): emission of a photon of exact energy  $\Delta E = hf$ . The same energy gaps appear in both processes.

This single idea — quantised energy levels — will explain the hydrogen spectrum numerically and lead us, by the end of the chapter, to Schrödinger's model of the atom.

## 2.7 Applications of spectra

- **Flame tests (practical work):** a platinum wire dipped in a metallic salt and held in a Bunsen flame colours it characteristically — sodium: intense yellow; potassium: lilac; calcium: brick red; copper(II): blue-green. This is the line emission spectrum seen by eye.
- **Street lamps:** sodium vapour lamps give an efficient yellow light dominated by the 589 nm doublet; mercury vapour lamps give a bluish-white light rich in green and violet lines (with

ultraviolet that requires a phosphor coating).

- **Neon advertising signs:** a discharge through neon at low pressure glows red-orange; other gases or coatings give other colours. The colour is the element's emission spectrum.
- **Astronomy:** the Fraunhofer absorption lines in starlight reveal the elements present in stellar atmospheres, their temperature, and even (through small wavelength shifts — Doppler effect) their motion toward or away from us.

#### WORKED EXAMPLE 2 Identifying a gas and finding a frequency

White light is shone through the cool vapour of an unknown element and then into a spectrometer. The continuous spectrum obtained is crossed by dark lines at 589.0 nm, 589.6 nm and several other wavelengths identical to those of sodium's emission spectrum. (a) Identify the element. (b) Calculate the frequency of the 589.0 nm line. (c) Explain why the lines are dark rather than bright.

#### SOLUTION

(a) The dark absorption lines coincide exactly with sodium's emission lines. Since each element has a unique line spectrum, the vapour is sodium.

(b) Using  $c = f\lambda$  with  $\lambda = 589.0 \text{ nm} = 5.890 \times 10^{-7} \text{ m}$ :

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m s}^{-1}}{5.890 \times 10^{-7} \text{ m}} = 5.09 \times 10^{14} \text{ Hz.}$$

(c) Sodium atoms absorb photons of these wavelengths strongly, then re-emit the energy in random directions. Very little of the re-emitted light travels forward into the spectrometer, so the beam is weakened at those wavelengths and the lines appear dark against the continuous background.

#### EXERCISE 2 Spectra around us

1. Classify each source by the type of spectrum it produces: (i) a candle flame made yellow by sodium impurities; (ii) the glowing filament of a torch bulb; (iii) a low-pressure mercury discharge tube; (iv) starlight crossing the star's cooler atmosphere.
2. A line in the hydrogen spectrum has wavelength 486 nm. Calculate its frequency and the energy of one photon in joules and in electron volts (use  $h = 6.63 \times 10^{-34} \text{ J s}$ ,  $e = 1.60 \times 10^{-19} \text{ C}$ ).
3. Explain why the absorption lines of a gas are found at exactly the same wavelengths as its emission lines.

**SOLUTION** Exercise 1

1. (i) Line emission spectrum (the sodium atoms in the hot, low-density flame emit their yellow doublet). (ii) Continuous spectrum (hot solid). (iii) Line emission spectrum (low-pressure gas in a discharge). (iv) Line absorption spectrum (continuous spectrum crossed by dark Fraunhofer lines).

2. Frequency:  $f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m s}^{-1}}{4.86 \times 10^{-7} \text{ m}} = 6.17 \times 10^{14} \text{ Hz}$ .

Photon energy in joules:  $E = hf = 6.63 \times 10^{-34} \text{ J s} \times 6.17 \times 10^{14} \text{ Hz} = 4.09 \times 10^{-19} \text{ J}$ .

Photon energy in eV:  $E = \frac{4.09 \times 10^{-19} \text{ J}}{1.60 \times 10^{-19} \text{ J eV}^{-1}} = 2.56 \text{ eV}$ .

3. An atom can only absorb the precise photon energies that carry it between its allowed energy levels, and these are exactly the energies it releases when it de-excites. Since  $E = hf = hc/\lambda$ , equal energy jumps mean equal wavelengths: the wavelengths a gas emits are precisely those it absorbs.

**KEY POINTS** Key points of this section

- A spectrometer (collimator + prism or grating + telescope) separates light into its wavelengths.
- Continuous spectrum: all wavelengths; from hot solids, liquids, dense gases.
- Line emission spectrum: discrete bright lines on a dark background; from low-pressure gases in discharge tubes or flames.
- Line absorption spectrum: continuous spectrum crossed by dark lines; produced when white light crosses a cooler gas; dark lines match the emission wavelengths exactly (Kirchhoff's 3rd law).
- Each element has a unique line spectrum — a fingerprint used in flame tests, street lamps, neon signs and the analysis of starlight.
- Line spectra prove that atoms exchange energy only in definite amounts: atomic energy is quantised.
- The electron volt ( $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$ ) is the practical energy unit for atomic transitions.

### 3 ENERGY LEVELS, THE ELECTRON VOLT AND THE EQUATION $E = hf = E_2 - E_1$

In the previous section we saw that a hot hydrogen gas emits light only at certain well-defined wavelengths, producing a line spectrum, and that a cool gas absorbs exactly those same wave-

lengths. This observation is incomprehensible if the electron in an atom could have any energy. It becomes perfectly natural if the electron can only occupy certain **allowed energies**. In this section we make that idea precise: we introduce **energy levels**, a practical unit of energy at the atomic scale (the **electron volt**), and the fundamental equation  $E = hf = E_2 - E_1$  that links every spectral line to a jump between two levels.

### 3.1 Quantisation of atomic energy

Imagine climbing a staircase in the dark: your foot can rest on a step, but never between two steps. The electron in an atom behaves in the same way: it can only possess certain definite values of energy, called **energy levels**.

#### DEFINITION Energy levels, ground state, excited states

The energy of an electron bound in an atom is **quantised**: it can only take certain discrete values  $E_1, E_2, E_3, \dots$  called energy levels. The lowest level  $E_1$  is the ground state; all higher levels are excited states. The atom is normally found in its ground state.

For the hydrogen atom, the allowed energies are given by the remarkably simple formula (which we shall justify in the next section):

$$E_n = -\frac{13.6}{n^2} \text{ eV}, \quad n = 1, 2, 3, \dots$$

where the integer  $n$  is called the **principal quantum number**. Note the crucial features:

- The energies are **negative**. This is not a defect: it expresses that the electron is bound to the nucleus.
- The levels get closer and closer together as  $n$  increases ( $E_1 = -13.6 \text{ eV}$ ,  $E_2 = -3.40 \text{ eV}$ ,  $E_3 = -1.51 \text{ eV}$ ,  $E_4 = -0.85 \text{ eV}$ , ...), and they accumulate towards  $E_\infty = 0$ .
- $E = 0$  corresponds to an electron just removed from the atom: the atom is **ionised**.

#### PROPERTY Meaning of the negative sign

The zero of energy is chosen for an electron at rest infinitely far from the nucleus (a free electron). A negative energy  $E_n < 0$  therefore means the electron is bound: one must supply energy  $|E_n|$  to free it. The value  $E = 0$  is the threshold of **ionisation**.

#### DEFINITION Ionisation energy

The ionisation energy of an atom is the minimum energy required to remove the electron completely from the ground state. For hydrogen:

$$E_{\text{ion}} = E_\infty - E_1 = 0 - (-13.6 \text{ eV}) = 13.6 \text{ eV}.$$

